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Effect of Fuel Injection Location on Combustion Instability in a Dump Combustor

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The acoustic behavior of a model combustor with backward facing step is studied, to examine the effect fuel injection location on the combustion instability. Six different injection locations, such as far upstream of the step, near upstream of the step, injection at the step, near downstream to the step, in the recirculation zone, and far downstream to the step, are investigated. Acoustic characteristics of the combustor over a wide range of equivalence ratios and Reynolds number based on step height are carried out. Liquefied petroleum gas (LPG) is used as fuel for the acoustic characterization. The effect of injection location on the blow off limits is presented, to identify specific conditions corresponding to unstable regions of combustion. Simultaneous unsteady pressure measurement, fuel flow fluctuations (by Mie scattering) and chemiluminescence imaging are done for conditions corresponding to local maxima and local minima of acoustic amplitude as the air flow rate is varied. Methane is used as fuel for these studies. The fuel injection location dictates the effect of fuel flow fluctuations (equivalence ratio fluctuations) on heat release and the corresponding acoustic oscillations. It is shown that within an optimum distance upstream of the step, the characteristics of vortex-shedding dominated acoustic excitation are predominated by those of the equivalence ratio fluctuations. Local maxima and minima of acoustic amplitude with air flow rate when fuel is injected downstream is from constructive and destructive interference of oscillations propagating along vertical and acoustic streamlines.

I. Introduction

Low NOx emission, noise reduction and fuel economy are the current focus of the gas turbine community. Operating combustors with lean mixtures are preferred on these grounds. Combustion of lean mixtures is prone to combustion instabilities. The primary concerns over combustion instability are the low frequency and high amplitude oscillations that lead to reduction in operating envelopes and hence performance, and structural failures. The complexity arises because of its interdisciplinary nature. The primary mechanism through which the combustion dynamics could be understood is the coupling between fluid dynamics, chemical heat release, and acoustics. Acoustics plays a major role by providing the feedback between flow and the unsteady chemical heat release. Through acoustic feedback, the vortex shedding in the flow modulates the unsteady heat release [1]. The heat release acts as a source of sound. In confined flows such as ducts, the location of the source of sound, i.e., the heat release, is one of the determining factors of combustion instability [2,3]. Vortical structures grow and detach in presence of external perturbations, causing the flame to flap as the vortices are shed past it.

The location of fuel injection relative to the flame holder is quite varied in gas turbines depending on the application. In land based gas turbines, the fuel is injected upstream of the flame stabilization zone to create a reasonably premixed mixture of reactants. In flight gas turbines, the fuel is injected downstream of the dump diffuser and the fuel and air are intensely mixed and burned within a short distance downstream. In either case, the location of fuel injection is within the acoustic field produced in the combustor. In the case of injection upstream of the flame stabilization zone, oscillatory injection of fuel in response to the prevalent acoustic field in the air inlet leads to equivalence ratio fluctuations – a prominent mechanism of combustion instability [2]. In this case, the distance between the fuel injection location and the flame stabilization zone dictates the delay of mixture fraction pockets being convected along the air flow. In the case of injection downstream of the dump diffuser, it has recently

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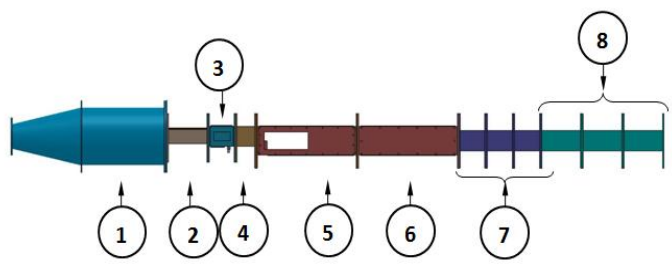
been shown that regions of constructive and destructive interference of axial velocity fluctuations between vortical and acoustic waves, the former propagating convectively and the latter at the speed of sound, could lead to zones of high fluctuations causing flame oscillations, alternating with 'silent' zones, depending upon the flow and geometric conditions [4,5].

The instabilities are also caused by interactions of combustor pressure oscillations with reactants supply rates, giving rise to equivalence ratio oscillations. These oscillations are convected at the speed of mean flow in combustor giving rise to large amplitude and heat release oscillations which drive the pressure oscillations [6]. Also when the fuel injection is done far upstream from the step, the equivalence ratio arriving at the flame becomes almost steady making the flame vortex interactions to become dominant. When the fuel is injected close to the step, the equivalence ratio arriving at the flame exhibits oscillations [7].

It is therefore important to study the effect of the location of fuel injection within the self-excited acoustic field relative to the flame stabilization/heat release rate fluctuation location on the combustion instability characteristics of a combustor. For this purpose, an attempt is made in the present work to investigate the combustion instability of a model half dump combustor, with LPG (liquefied petroleum gas) as fuel. The case of fuel injection at the dump plane (with methane as fuel) has already been reported earlier [8,9]. The experiments are done for various fuel injection locations with respect to the flame holder (both upstream and downstream) and combustor lengths. Wide ranges of air and fuel flow rates are considered.



1. Settling chamber ; 2. Inlet duct; 3. IOAM; 4. Upstream extension ducts (75mm); 5. Test section; 6. Big extension duct (370mm); 7. Downstream extension ducts (100mm); 8. Downstream extension ducts (150mm)

Figure 1. Schematic of combustor test rig

II. Experimental Details

A. Experimental Test Rig

The test rig is intended to carry out experiments on a model combustor, with backward facing step as the flame holder. The experimental setup consists of settling chamber, inlet duct, IOAM (inlet access optical module), upstream extension duct (75mm), test section, long extension duct (370mm) and downstream extension ducts of length 100mm & 150mm. The schematic of the test setup is shown in Figure 1. The purpose of the settling chamber is to decouple the pressure fluctuations in the

Table : 1 Summary of combustor configurations

S.No	Configuration	Details	Overall Lengths (mm)
1	Far upstream (Xf = 175mm)	Fuel injection from IOAM (Upstream of the step), with injection location of 175 mm from the Inlet duct entry.	1365,1515,1665,1815
2	Near upstream (Xf = 225mm)	Fuel injection from IOAM (Upstream of the step), with injection location 225 mm from the Inlet duct entry.	1365,1515,1665,1815
3	At the step (Xf = 365mm)	Fuel injection from Combustor (Through the step).	1365,1515,1665,1815
4	Near downstream (Xf = 385mm)	Fuel injection from Combustor (Downstream of the step), with injection location 60mm (2H) from the step, at a height of 30mm (H) from the floor.	1365,1515,1665,1815
5	Recirculation zone (Xf = 385mm)	Fuel injection from Combustor (Downstream of the step), with injection location 60mm (2H) from the step, at a height of 15mm (H/2) from the floor.	1365,1515,1665,1815
6	Far downstream (Xf = 445mm)	Fuel injection from Combustor (Downstream of the step), with injection location 120mm (4H) from the step, at a height of 15mm (H/2) from the floor.	1365,1515,1665,1815

delivery line to the entry of the test setup. The exit of the settling chamber contracts abruptly with a contraction ratio of 17.45:1. This is done to have an acoustically open boundary condition at the entry of the inlet duct. The purpose of the inlet duct is to generate near uniform flow in the combustor entry. This is achieved by using stacked tubes and meshes of sizes 600,200 and 140 µm. Along with flow straightening, these structures also act as flame arresters, in case of premixed combustion studies. The purpose of the IOAM is to provide optical access to study combustor inlet flow and the fluctuations in the fuel flow when injected upstream. It contains windows of at top and bottom for laser sheet access and on the sides for viewing/imaging. The fuel is injected from the bottom wall of the IOAM for partially premixed combustion studies. The test section has a backward facing step of 30 mm height. Downstream of the step, the cross section of the test section expands to 60 x 60 mm. Optical access is provided in all four sides of the test section. The top and bottom windows are for laser sheet access. Magnesium oxide is used as seeding particles for fuel flow visualization and to quantify its temporal fluctuations.

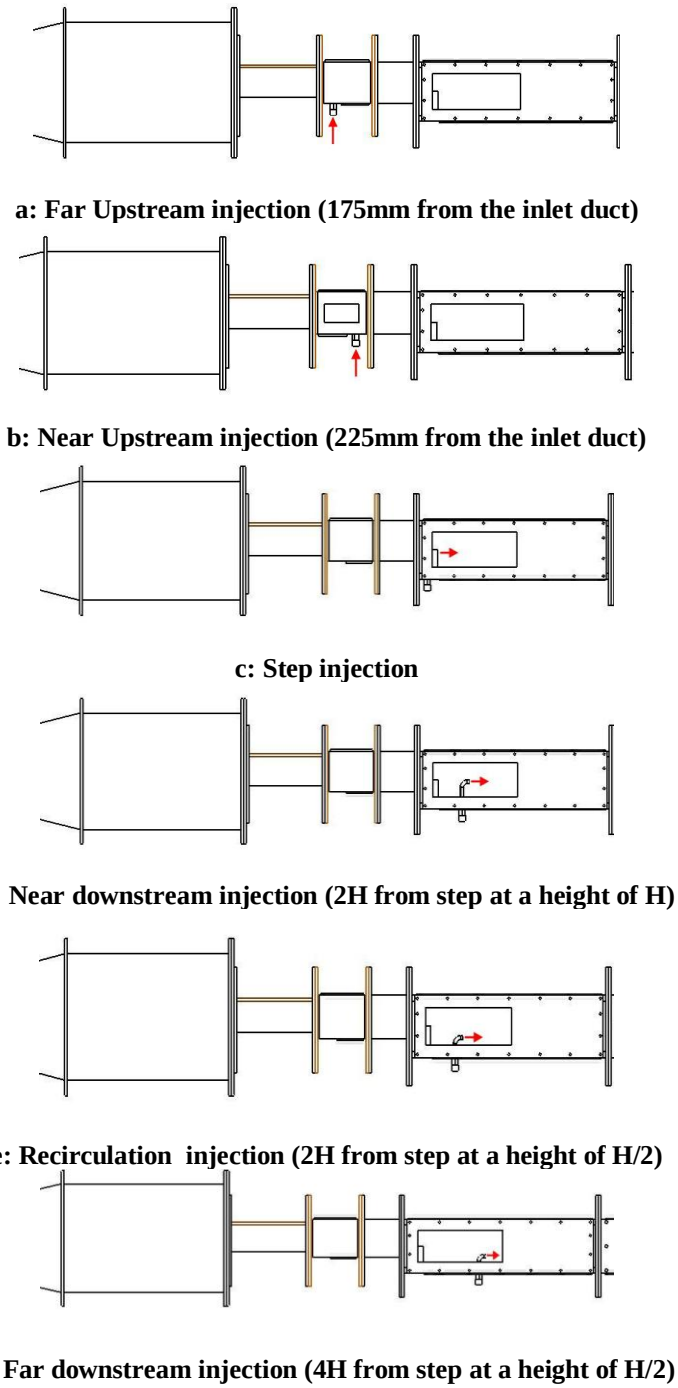


Figure 2. Different injection schemes.

B. Measurements

The acoustic pressure (unsteady pressure) measurements are done using the PCB make piezoelectric transducers mounted at various locations along the combustor length. Data acquisition is done using National Instruments analog-to-digital card and LabView software. Optical diagnostics is done simultaneously for fuel flow fluctuation and chemiluminescence. High speed charge-coupled device (CCD) cameras of frame rate 2250 are used. The seeded fuel is illuminated with 514 nm argon-ion laser. Sheet optics is constructed using concave lens, convex lens and cylindrical lens to achieve laser sheet. The camera recording fuel flow fluctuation is fitted with 514 nm filter. The camera recording chemiluminescence is fitted with 431 nm filter.

C. Configuration Details

The crucial parameters in the study of acoustic characterization of the backward facing step combustor are the total length L, the location of the step X_h with respect to overall length and the location of the fuel injection X_f with respect to the flame holder location. Different test configurations are achieved by varying these parameters. In the present study, the distance of the step from the inlet is not varied. The overall length and fuel injection schemes are varied as described below.

1. Fuel injection location: Six different locations are chosen, as explained in Table 1. The schematic of the injection schemes are shown in Figure 2. The first two configurations are upstream fuel injection, which are more premixed when the fuel reaches the flame holder, than the other cases correspond to non-premixed modes of combustion.

2. Overall length L: The base configuration is constructed with settling chamber, inlet duct (150 mm long), IOAM (100 mm long), upstream extension duct (75 mm long), combustor (370 mm long), long extension duct (370 mm long) and three downstream extension ducts (100 mm long). The length of the base configuration is 1365 mm. By adding three more extension ducts of length 150 mm each, the overall lengths of 1515 mm, 1665 mm and 1815 mm are achieved.

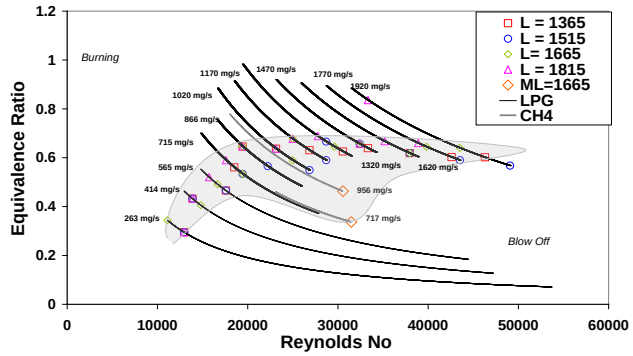
III. Results and Discussion

A. Blowoff Limits

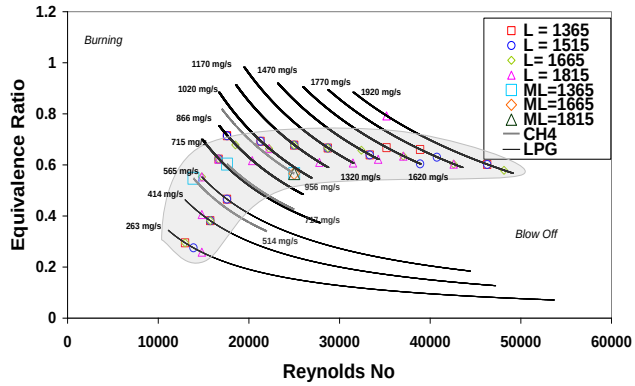
The blow-off limits for every length and fuel flow rate of each configuration are shown in Fig. 3 in the equivalence ratio versus Reynolds number domain. Hyperbolae in this domain correspond to constant fuel flow rate. Lower hyperbolae correspond to lower fuel flow rates and vice versa. In each of these graphs, as one traverses a constant fuel flow rate line from the top left corner, one encounters a data point for the given length of the combustor when the flow blows off. So, the bottom right part of the graphs corresponds to the blow-off condition.

In general, the pattern of the blowoff limits form a pattern of an inverted 'L' shape, i.e., for low fuel flow rates, a narrow critical range of Reynolds number exists beyond which the flame blows off over a wide range of low equivalence ratio, whereas for high fuel flow rates, a narrow critical range of equivalence ratio exists below which the flame blows off over a wide range of high Reynolds number.

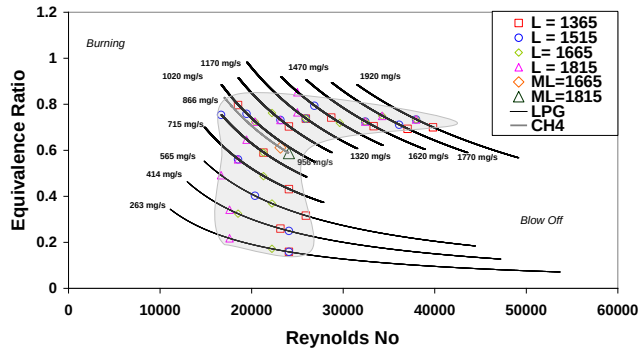
This general pattern is significantly altered by the



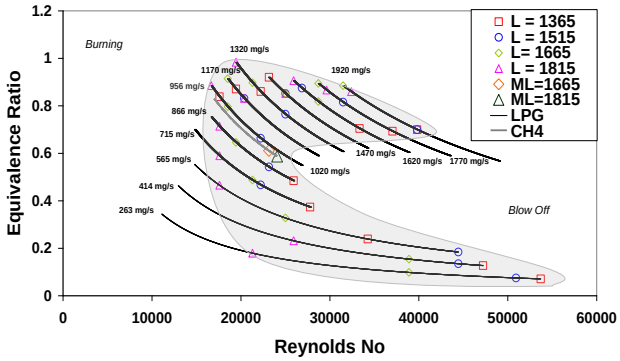
(a) Far upstream injection



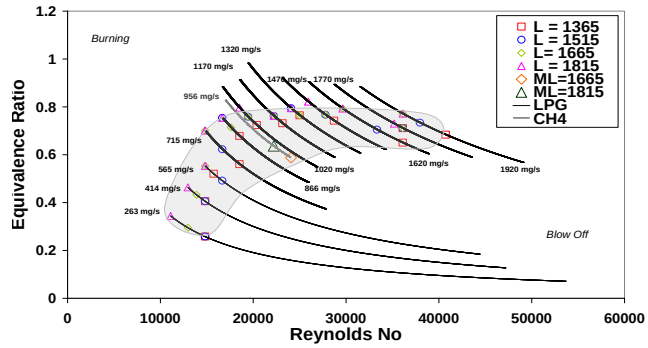
(b) Near upstream injection



(c) Near downstream injection

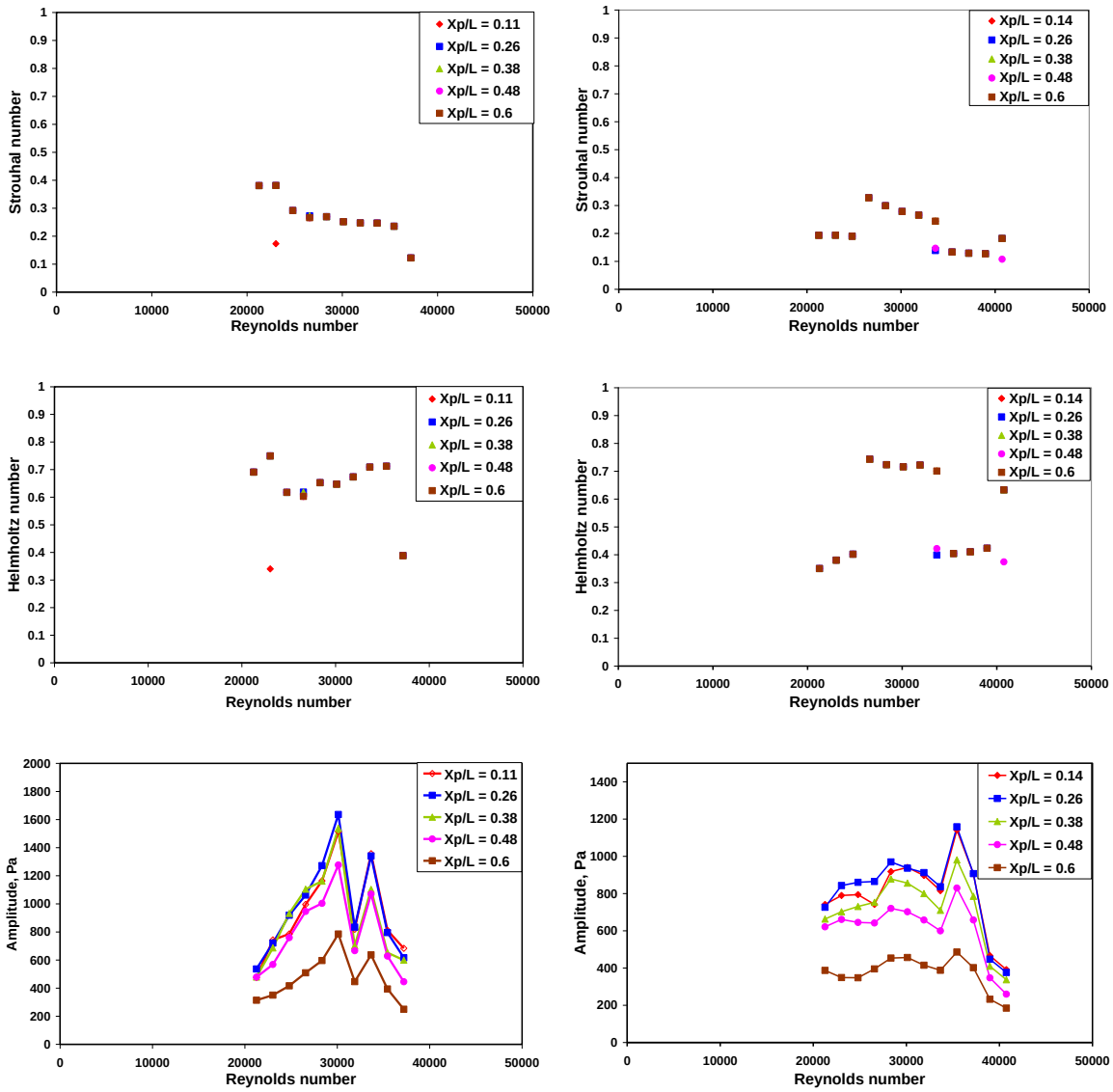


(d) Recirculation injection



(e) Far downstream injection

Figure 3. Blow-off limits.



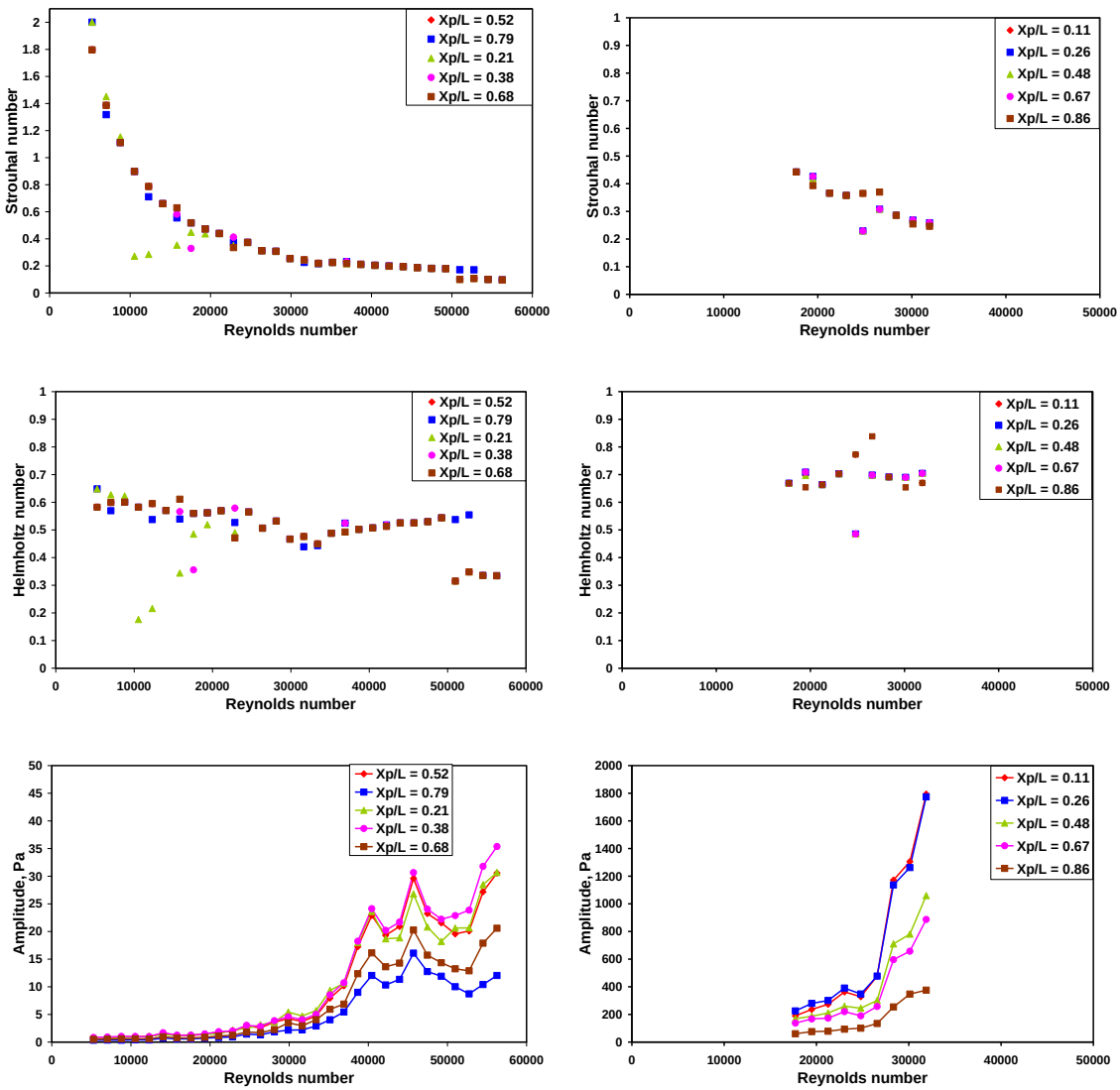
(a) Far upstream injection **(b) Near upstream injection**

Figure 4. Acoustic characteristics at different injection locations.

choice of fuel injection location. In the case of the far upstream injection (Fig. 3a), the bottom leg of the inverted 'L' is nearly truncated, it is slightly developed for the case of near upstream injection (Fig. 3b), quite prominent for near downstream injection (Fig. 3c), very elongated along the hyperbolae for the recirculation zone injection (Fig. 3d), and again shrunk back for the far downstream injection case (Fig. 3e). The existence of the bottom leg of the inverted 'L' signifies the extent of non-premixedness that supports the combustion even at low equivalence ratios up to a critical Reynolds number. This explains the near absence to a slight presence of this region of blowoff limits in the upstream injection cases, respectively far and near the step. In the downstream injection cases, high flow velocity causes the flame to stand out from the injection point, leading to some premixing, and hence destabilization of the flame. This effect is the least in the recirculation zone and the most far downstream, which explains the observed maps for the downstream injection cases.

B. Acoustic Characterization

To understand the acoustic behaviour of the combustor, unsteady pressure data is acquired with increasing the air flow rate, keeping the fuel flow rate constant. By this the Reynolds number and equivalence ratio are varied. From



(c) Step injection **(d) Near downstream injection**

Figure 4 (contd.). Acoustic characteristics at different injection locations.

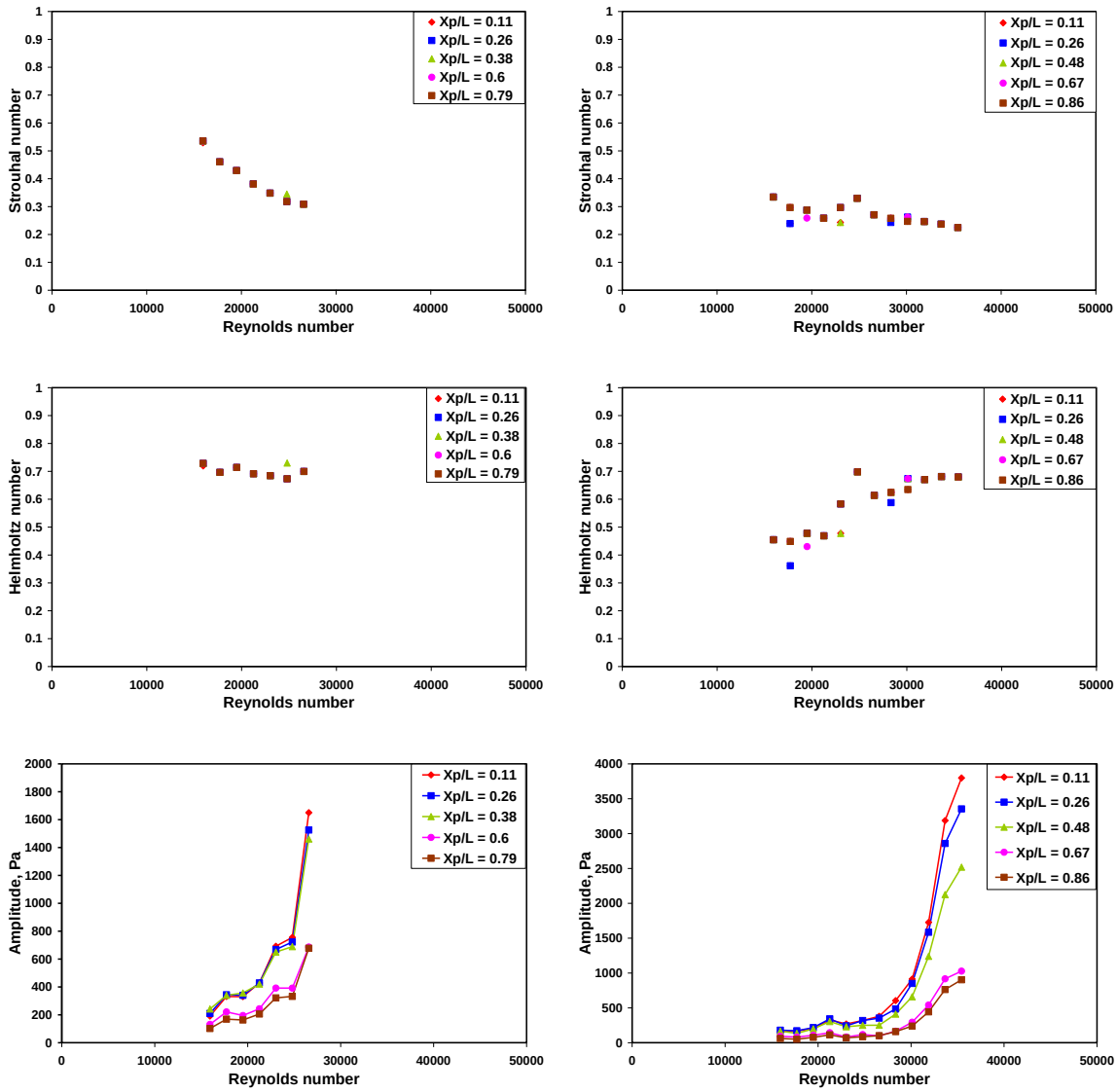
the unsteady pressure data, the dominant frequency and its amplitude for various Reynolds number are plotted. The dimensionless quantities Helmholtz number and Strouhal number are used to characterise the acoustic behaviour of the c combustor in relation to the vortex shedding process. The Helmholtz number and Strouhal number are as defined below.

$$\text{Helmholtz number} \quad He = f\,L\,/\,c$$

$$\text{Strouhal number} \quad St = f\,h\,/\,u$$

where, f – dominant frequency of the signal, L – overall combustor length (m), c – speed of sound (m/s), h – flame holder major dimension (step height), u – combustor entry velocity (m/s)

1. Far upstream injection: In this case, the variation of Strouhal number, Helmholtz number and amplitude with respect to Reynolds number, is shown in Fig. 4(a). The Helmholtz number increases linearly with respect to Reynolds number, near Re= 30000, indicating a vortex shedding mode. Thus, coupling between vortex roll up and



(e) Recirculation injection **(f) Far downstream injection**

Figure 4 (contd.). Acoustic characteristics at different injection locations.

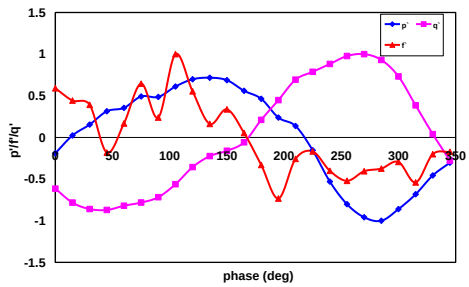
heat release predominantly prevails in this configuration, as observed by Altay et al. [7]. The amplitude variation has a minimum in the middle. This suggests that even though vortex mode predominates, the fluctuations in the fuel injected far upstream modulate the amplitude. Since the fuel flow fluctuation is carried by convection, the phase at which the fuel arrives the flame results in modulating the amplitude.

2. Near upstream injection: In this case, the variation of Strouhal number, Helmholtz number and amplitude with respect to Reynolds number, is shown in Figure 4(b). It can be seen from the Helmholtz number is constant with Reynolds number near Re = 30000, indicating the natural acoustic mode. The high amplitudes in this case, then may be due to near upstream injection, which overrides the vortex shedding fluctuating the heat release. Thus, in this case fuel flow fluctuation may be dominating over the vortex roll up, in modulating the heat release. This is the opposite of the conclusion arrived by Altay et al. [7], who did not inject so close to the step as in the present case.

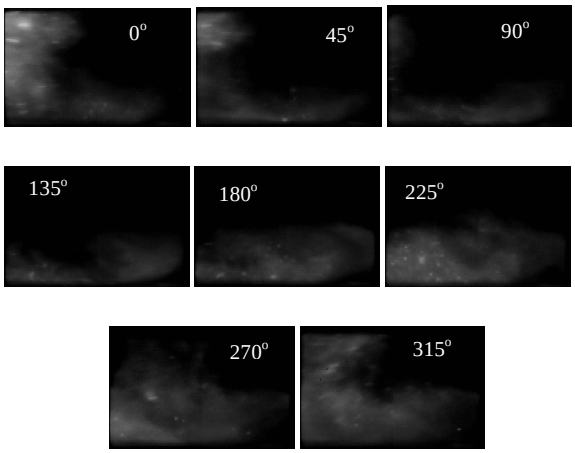
3. Step injection: In this case, the variation of Strouhal number, Helmholtz number and amplitude with respect to Reynolds number is shown in Figure 4(c). It can be seen that the amplitude increases slowly but the levels are not high. At Re = 40000 and above, the amplitude starts to rise and the Helmholtz number increases linearly, as in

[8,9]. The range of operation in terms of Reynolds number is higher as compared to other cases, which is due to the non-premixedness of the reactants at the step.

4. Downstream Injection: In these cases, the variation of Strouhal number, Helmholtz number and amplitude with respect to Reynolds number, are shown in Figs. 3(d), 3(e) and 3(f). Since the fuel injection is downstream of the flame holder, the dominance of fuel flow fluctuations over the vortex and heat release interaction is expected to be lesser. Also, in these cases, the amplitudes are considerably higher compared to upstream injection cases. In these cases it could be seen that, both acoustic and vortex modes are present. The high amplitudes are observed during mode sifts between acoustic mode and vortex mode or vice versa.

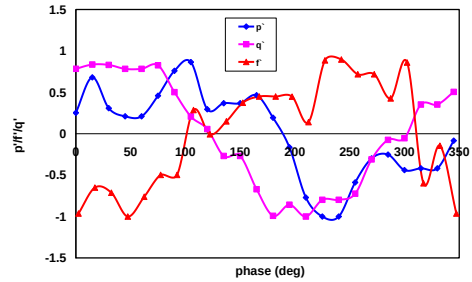


(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time history

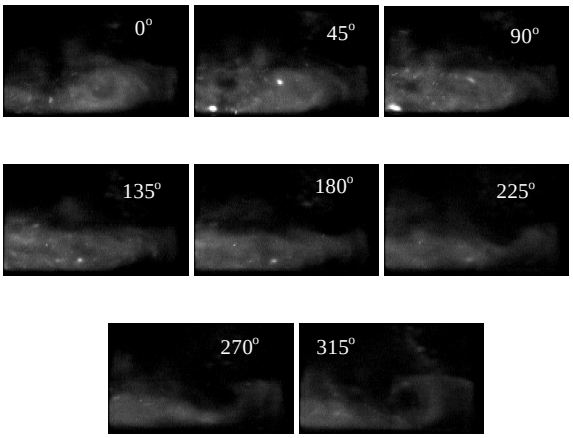


(b) Instantaneous chemiluminescence images

Figure 5. Far upstream injection (local maxima).



(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time history



(b) Instantaneous chemiluminescence images

Figure 6. Far upstream injection (local minima).

C. Simultaneous Fuel Injection Fluctuations and Flame chemiluminescence Imaging

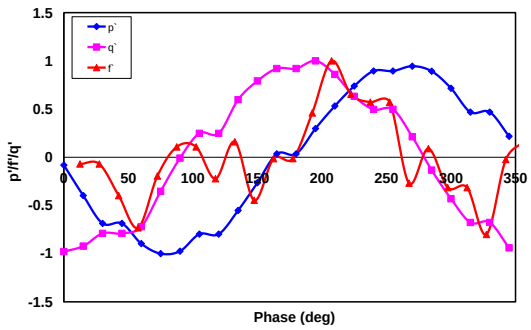
1. Far upstream injection: Figure 5(a) illustrates the phase differences between the pressure fluctuation, fuel flow fluctuations and the heat release fluctuations for local maxima of far upstream injection case. The corresponding fuel flow rate is 956mg/s and Reynolds number is 21258. The phase difference graph is plotted for one cycle of the dominant frequency, which is 43.9Hz. It is seen that the phase averaged plot has a good correlation with the instantaneous images. The phases in the phase diagram are all calculated at the step. Initially from 0□ phase, the flame is pushed out due to reduction in fuel flow, leading to a reduced heat release rate. The vortex roll-up starts at around the phase of 90□ and the heat release fluctuation tends to increase. Towards the end of the cycle the fuel flow fluctuations shows an increasing trend due to which the flame flashes back.

Figure 6(a) illustrates the phase differences between the pressure fluctuation, fuel flow fluctuations and the heat release fluctuations for local minima of far upstream injection case. The corresponding Reynolds number is 21258 and acoustic dominant frequency is 111.62 Hz. It's seen that, at around 225□ phase, vortex rolls up and the fuel flow fluctuations show an increasing trend. Following this due to the decrease in the fuel flow, flame could not flash back, even though the vortex continues to roll-up. This behavior is different from the far upstream local maxima case and here the pressure amplitudes are lower than the previous case. When compared to the previous case the fuel flow fluctuations are 180□ out of phase with the chemiluminescence fluctuations. This leads to the drop in the pressure amplitude even though the pressure p' and the heat release q' fluctuations are in phase with each other.

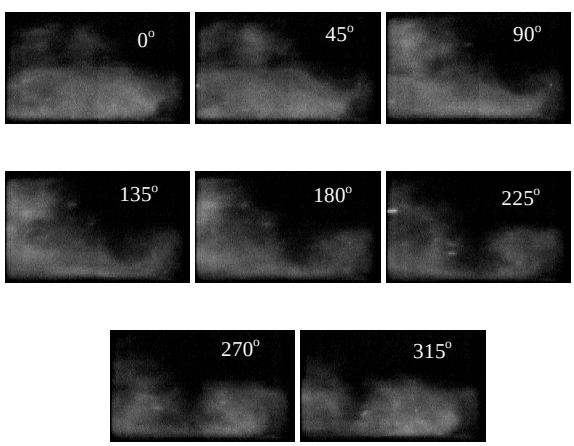
Here, the flame actually entrains the vortex which is already rolled up, so the change in the flame area and the heat release fluctuations are not significant.

2. Near upstream injection: The phase diagrams for the fuel flow, pressure and the heat release fluctuations are shown with the corresponding instantaneous chemiluminescence images at the regular intervals in the Figs. 7(a) and (b). The dominant mode is 118.78Hz. The corresponding Reynolds number is 19487 and fuel flow rate is 956mg/s. Near 45□ phase the flame tries to move upstream but the equivalence ratio fluctuations are not favorable. Even though the flame is stretching, lower fuel rich mixture availability reduces the heat release. The vortex rolls-up at around 135□ phase curls the flame, increases the flame surface area and the fuel fluctuations are also favorable, leading to a higher heat release rates. But at the phase when the fuel flow fluctuation starts to increase, the vortex roll up becomes stronger and tends to pull the flame back. This prevents the flame from flashing back, as it is seen in the image at 225□ phase.

For the local minima of near upstream case, the Reynolds number is 21258 and the fuel flow rate is 956mg/s. The acoustic dominant frequency is 120 Hz. Phase diagram and chemiluminescence images for this case are shown in Figs. 8(a) and (b). In this case at 0□ phase the flame tends to move upstream. The vortex roll-up and the fuel flow fluctuations going negative pushes off this flame downstream. The vortex rolls up at around 90□ phase at which the fuel flow fluctuations show a negative trend leading to a reduced heat release. At this point with further vortex roll up the heat release is found to increase. At a phase of 315□ the vortex seems to detach, the flame tends to rise up and tries to move ahead. This is a consequence of the fuel flow fluctuation being positive.

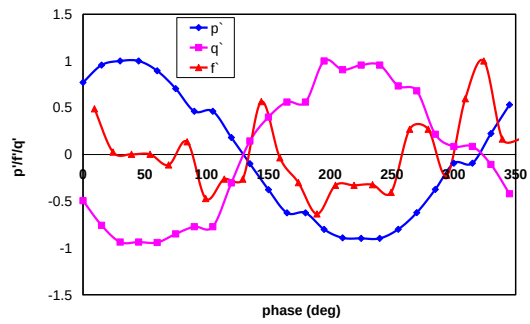


(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time history

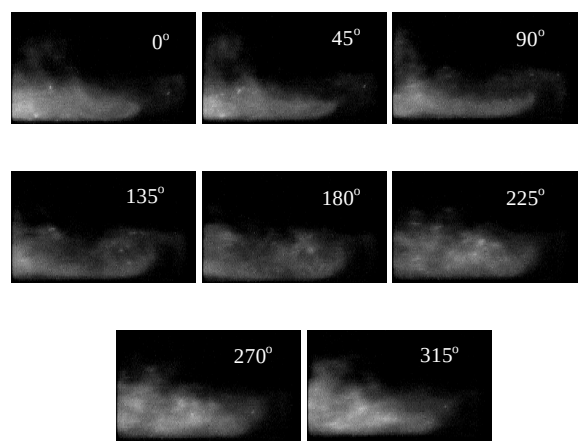


(b) Instantaneous chemiluminescence images

Figure 7. Near upstream injection (local maxima).



(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time history

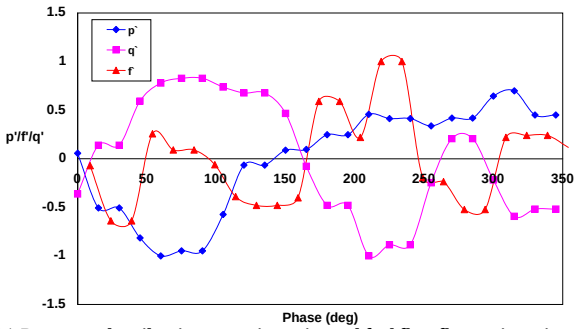


(b) Instantaneous chemiluminescence images

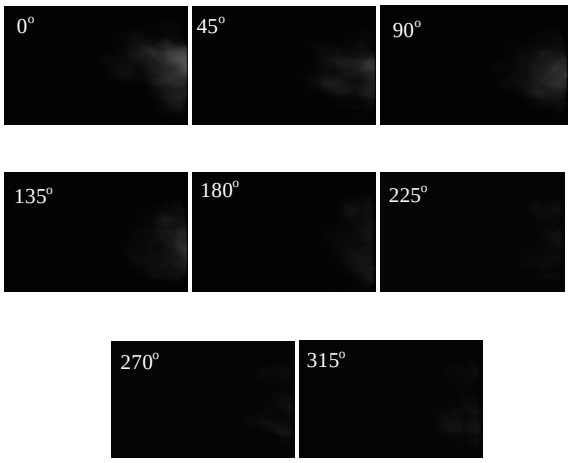
Figure 8. Near upstream injection (local minima).

3. Near downstream injection: The phase diagrams for the fuel flow, pressure and the heat release fluctuations are shown with the corresponding instantaneous chemiluminescence images for local minima are shown in Figures. 9(a) and (b). The dominant mode is 132.21Hz and the corresponding Reynolds number is 17715 and fuel flow rate is 956mg/s. For the local maxima, the dominant mode is 158.80Hz and the Reynolds number is 19487. Figures. 10(a) and (b) correspond to the local maxima. All three fluctuations (p', q' & f') are appear to be 180 deg out of

phase between local maxima and local minima. Also for local maxima the heat release rate, fuel flow fluctuations and the pressure are in phase with each other, which is not seen in the case of local minima. This fluctuation happens due to some premixing which leads to an unstable flame.

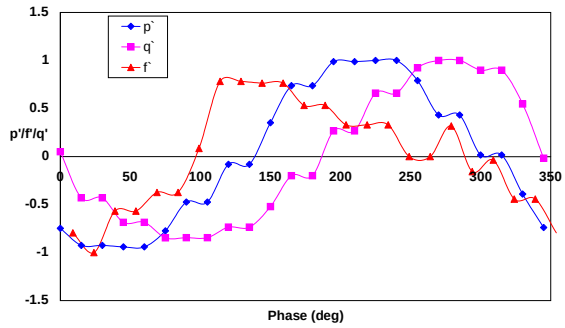


(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time history

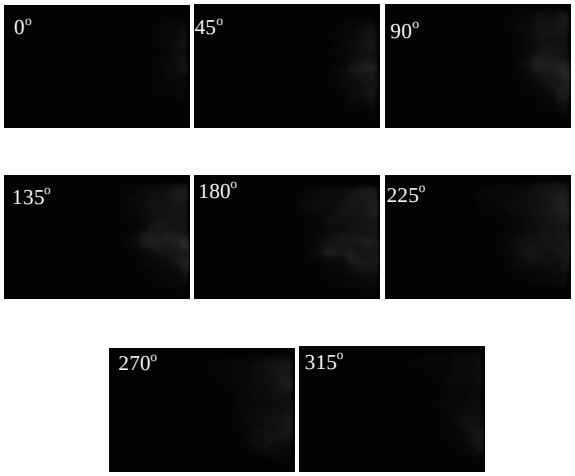


(b) Instantaneous chemiluminescence images

Figure 9. Near downstream injection (local minima).



(a) Pressure, chemiluminescence intensity and fuel flow fluctuations time History



(b) Instantaneous chemiluminescence images

Figure 10. Near downstream injection (local maxima).

IV. Conclusion

A systematic study on role of fuel injection location over combustion instability of a model combustor with backward facing step as flame holder is carried out. Fuel injection is done upstream to the step, through the step and downstream of the step. Two injection location upstream to the step and three injection locations downstream to the step are identified. The study is carried out in two phases. In the first phase the unsteady pressure measurements are done for different combustor configurations (based on total length and fuel injection scheme) over a wide range of inlet Reynolds number and equivalence ratios. LPG is used in this phase. Based on the acoustic data, Blow off limits is identified. Blow off limits provides information on the operating equivalence ratio regions of the combustor and effect of configuration on the variation of blow off equivalence ratio. Blow off limit maps throws light on the effect of fuel injection locations on stability of the combustor. Recirculation and near downstream injection cases exhibit stable flame near low Reynolds numbers, indicating the extent of unmixedness. Far upstream and far downstream injection cases are considerably unstable. The optimum injection location for stable flame may be located close to near upstream to the step. Near upstream injection case exhibits acoustic mode with Reynolds number, indicating the dominance of fuel flow fluctuations over flame vortex interactions. In other cases vortex mode is present with modulation by the fuel flow fluctuations. In downstream injection cases both vortex and acoustic modes are present.

The second phase is to carry out simultaneous acoustic measurement, fuel flow fluctuations (Mie scattering) and chemiluminescence for specific flow conditions and configuration. Conditions corresponding to Local minima and local maxima of acoustic amplitude are chosen. Methane is used as fuel in the second phase. The instantaneous

chemiluminescence and phase averaged pressure, flue flow and chemiluminescence are correlating well. These studied brings out phase relation between above three parameters. It is clearly seen that the phase of fuel flow fluctuations plays a major role in deciding the behavior of the combustor. For conditions corresponding to local minima, for some of the cases the availability of fuel being less, the heat release is lower even though the vortex roll up is favorable. The fuel injection location dictates the phase relation between fuel supply and heat release, thus modulating the instability behavior of the combustor.

In practical cases the location of the fuel injector in combustors are most important as far as the stability of the combustor is concerned. The upstream injection cases show that the fuel flow fluctuations play a vital role in a stable flame zone. The discussion implies that the near upstream injection leads to a stable flame zone, which can also be seen from the blow off limits. This is useful for the practical combustors to locate the injector at the upstream of the flame stabilization so as to maintain the premixedness. The location of the injector should not be too near to the flame stabilization zone leading to a non premixed flame and high flame vortex interactions. So an optimum distance, which leads to less noise at a lean equivalence ratio, should be opted for stable combustor conditions.

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